

ECE 647: Lecture 14

Transformation to a Convex Problem (Continued)

Examples

(3) Equality constraints can sometimes be converted into inequality constraints, and vice versa, if we know additional structure of the problem.

$$\begin{array}{ll} \min_{x_1, x_2} & x_1 + x_2 \\ \text{s.t.} & x_1^2 + x_2^2 = 1 \text{ **non-linear**} \\ & \Updownarrow \\ \min_{x_1, x_2} & x_1 + x_2 \\ \text{s.t.} & x_1^2 + x_2^2 \leq 1 \text{ **convex**} \end{array}$$

We can make the following conversion to the inequality constraints:

$$\begin{array}{ll} \min_x & f_0(x) \\ \text{s.t.} & h_i(x) = 0 \\ & \Updownarrow \\ \min_x & f_0(x) \\ \text{s.t.} & h_i(x) \leq 0 \end{array}$$

when we know the optimal point must lie on the boundary of the new constraints set.

(4) Implicit constraints can be made explicit

Suppose

$$f(x) = \begin{cases} x^\top x & \text{if } Ax = b \\ +\infty & \text{otherwise} \end{cases}$$

Then

$$\begin{aligned} \min_x f(x) &\Leftrightarrow \min_x x^\top x \\ \text{s.t. } Ax &= b \end{aligned}$$

(5) Simplify objective/constraint functions and introduce additional constraints

$$\begin{aligned} \min_x f_0(A_0x + b_0) \\ \text{s.t. } f_i(A_ix + b_i) &\leq 0 \\ h_i(x) &= 0 \\ \Updownarrow \\ \min_x f_0(y_0) \\ \text{s.t. } f_i(y_i) &\leq 0 \\ A_ix + b_i &= y_i, i = 0, 1, \dots, p \\ h_i(x) &= 0 \end{aligned}$$

- Sometimes works with nonlinear mappings:

$$\begin{aligned} \min_x f_0(g(x)) \\ \Updownarrow \\ \min_x f_0(y) \\ \text{s.t. } g(x) &= y \\ \Updownarrow \text{ if } f_0 \text{ is increasing, } g \text{ is convex} \\ \min_x f_0(y) \\ \text{s.t. } g(x) &\leq y \end{aligned}$$

Since f_0 is increasing, we want the smallest y . Drive y until $y = g(x)$ and $g(x)$ can no longer decrease.

- Conversely, equality constraints can be absorbed into the function

sometimes:

$$\begin{aligned}
& \min_x f_0(x) \\
& \text{s.t. } f_i(x) \leq 0 \\
& \quad Ax = b \\
& \Rightarrow \text{write this as } x = g(x) = F(z) + x_0, \text{ where } AF(z) = b \text{ and } Ax_0 = 0 \\
& \quad \Updownarrow \\
& \min_x f_0(F(z) + x_0) \\
& \text{s.t. } g(F(z) + x_0) = y
\end{aligned}$$

(6) Epigraph form

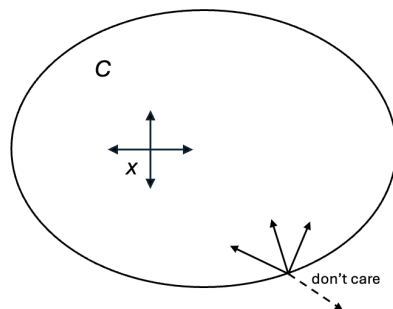
$$\begin{aligned}
& \min_x f_0(x) \\
& \text{s.t. } f_i(x) \leq 0 \\
& \quad \Updownarrow \\
& \min_x t \leftarrow \text{linear objective} \\
& \text{s.t. } f_0(x) - t \leq 0 \leftarrow \text{forces } f_0(x) = t \\
& \quad f_i(x) \leq 0
\end{aligned}$$

(7) Slack Variables (convert inequality constraints to equality constraints)

$$\begin{aligned}
& \min_x f_0(x) \\
& \text{s.t. } f_i(x) \leq 0 \\
& \quad \Updownarrow \text{ if } f_i \text{ is linear} \\
& \min_x f_0(x) \\
& \text{s.t. } f_i(x) + s_i = 0 \\
& \quad s_i \geq 0
\end{aligned}$$

Conditions for Optimality

When is a point x optimal? Roughly speaking, x is optimal when the function is non-decreasing in any direction pointing towards the interior of C .



- Define the direction derivative of a function f at $\bar{x}$ in a direction $d \in E$ as

$$f'(\bar{x}; d) = \lim_{t \rightarrow 0} \frac{f(\bar{x} + td) - f(\bar{x})}{t},$$

when this limit exists.

- If $f'(\bar{x}; d) = \langle a, d \rangle$ for all direction d , then the function is differentiable at $\bar{x}$, and the derivative is $\nabla f(\bar{x}) = a$